HABIB UNIVERSITY

Final Exam (SOLUTION)
Electromagnetic Theory
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Points: 100

Q1: [**30**] Consider an infinitely long, straight cylindrical conductor with radius a, carrying a steady current I. The current is non-uniformly distributed such that the current density $\mathbf{J}$ is proportional to the radial distance ρ from the z-axis. Assume $\mathbf{J} = K\rho\,\mathbf{a}_z$ for $\rho \leq a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

(a) [**5**] Given a and I, what is the value of K?

(b) [**5+5**] Using Ampere's law, determine the magnetic field intensity $\mathbf{H}$ both **inside** and **outside** the conductor.

(c) [**5**] Plot the magnitude of $\mathbf{H}$ to show its variation as a function of ρ.

(d) [**5+5**] Find the curl of $\mathbf{H}$ to establish that

$$\nabla \times \mathbf{H} = \mathbf{J}$$

is true **everywhere**, i.e., both inside and outside the cable.

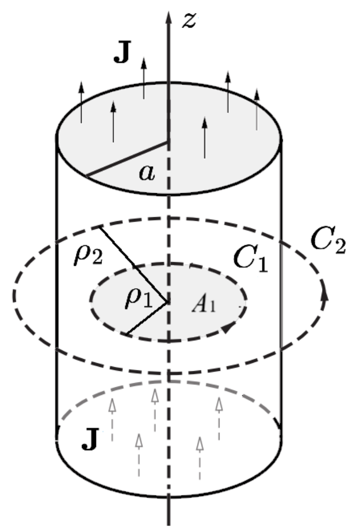

Figure 1: When using Ampere's law, one can consider the circular Ampere contours C_1 and C_2 with radii ρ_1 and ρ_2, respectively, where $0 < \rho_1 < a$ and $a < \rho_2 < \infty$, to calculate the magnetic field intensity both inside and outside the conductor.

Solution: We use the cylindrical coordinate system with the current I flowing in the z direction. Given the cylindrical symmetry, using Ampère's law to find $\mathbf{B}$ is convenient. According to the Biot-Savart law and the right-hand rule, $\mathbf{H}$ is in the $\hat{\boldsymbol{\phi}}$ direction, so that we have $\mathbf{H}(\rho) = H(\rho)\mathbf{a}_\phi$. We take two circular contours, C_1 and C_2, respectively, with radii $\rho_1 < a$ and $\rho_2 > a$, as shown in Figure 1. Applying Ampère's law along path C_1 lying inside the conductor, we have

$$\oint_{C_1} \mathbf{H} \cdot d\mathbf{l} = \int_{A_1} \mathbf{J} \cdot d\mathbf{s}$$

$$\int_{\phi=0}^{2\pi} (H_\phi \mathbf{a}_\phi) \cdot (\mathbf{a}_\phi \rho_1 d\phi) = \int_0^{2\pi}\int_0^{\rho_1} K\rho\mathbf{a}_z \cdot \mathbf{a}_z \rho d\rho d\phi = I_1$$

where I_1 is the portion of the total current I through the area $A_1 = \pi\rho_1^2$ enclosed by contour C_1. When $\rho_1 = a$, $I_1 = I$, this gives

$$\int_0^{2\pi}\int_0^{a} K\rho^2 d\rho d\phi = I \quad \Rightarrow \quad K = \frac{3I}{2\pi a^3}$$

From the above we have

$$H_\phi \rho_1 \int_0^{2\pi} d\phi = 2\pi K \frac{\rho_1^3}{3} \quad \Rightarrow \quad H_\phi = \frac{K\rho_1^2}{3}, \quad \rho_1 \leq a$$

Similarly, by applying Ampère's law along contour C_2 lying outside the conductor (i.e., $\rho_2 > a$), we find

$$H_\phi (2\pi\rho_2) = I \quad \Rightarrow \quad H_\phi = \frac{I}{2\pi\rho_2}, \quad \rho_2 > a$$

In summary, the $\mathbf{H}$ field at any position ρ is

$$\mathbf{H} = \begin{cases} \mathbf{a}_\phi \dfrac{I\rho^2}{2\pi a^3} & \rho \le a \\ \\ \mathbf{a}_\phi \dfrac{I}{2\pi\rho} & \rho > a \end{cases}$$

Since $\mathbf{H}$ is expressed in cylindrical coordinates, we use the cylindrical coordinate expression for curl. Note that the only nonzero component of $\mathbf{H}$ is H_ϕ, which varies only with ρ. Thus, the curl of $\mathbf{H}$ is

$$\nabla \times \mathbf{H} = \mathbf{a}_z \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho H_\phi) = \mathbf{a}_z \frac{1}{\rho}\left[H_\phi + \rho\frac{\partial H_\phi}{\partial\rho}\right]$$

so that for $\rho \le a$, we have

$$\nabla \times \mathbf{H} = \mathbf{a}_z \frac{1}{\rho}\left[\frac{I\rho^2}{2\pi a^3} + \rho\frac{2I\rho}{2\pi a^3}\right] = \mathbf{a}_z \frac{3I\rho}{2\pi a^3} = \mathbf{J}$$

Similarly, for $\rho > a$, we have

$$\nabla \times \mathbf{H} = \mathbf{a}_z \frac{1}{\rho}\left[\frac{I}{2\pi\rho} - \rho\frac{I}{2\pi\rho^2}\right] = \mathbf{0}$$

as expected, since $\mathbf{J} = 0$ for $\rho > a$.

Q2: [**25**] Consider an infinitely long, straight cylindrical conductor with radius a, carrying a steady current I. The current is non-uniformly distributed such that the current density $\mathbf{J}$ is proportional to the square root of the radial distance ρ from the z-axis. Assume $\mathbf{J} = K\sqrt{\rho}\,\mathbf{a}_z$ for $\rho \le a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

- **(a)** [**5**] Given a and I, what is the value of K?
- **(b)** [**5+5**] Find $\mathbf{A}$ inside the conductor ($\rho < a$).
- **(c)** [**5+5**] Find $\mathbf{A}$ outside the conductor ($\rho > a$).

Hint: Obtain $\mathbf{B}$ using Ampere's law or Biot-Savart's law, and solve $\mathbf{B} = \nabla \times \mathbf{A}$ to find the magnetic vector potential $\mathbf{A}$.

Solution: We use the cylindrical coordinate system with the current I flowing in the z direction. Given the cylindrical symmetry, using Ampère's law to find $\mathbf{B}$ is convenient. According to the Biot-Savart law and the right-hand rule, $\mathbf{H}$ is in the $\hat{\boldsymbol{\phi}}$ direction, so that we have $\mathbf{H}(\rho) = H(\rho)\mathbf{a}_\phi$. We take two circular contours, C_1 and C_2, respectively, with radii $\rho_1 < a$ and $\rho_2 > a$. Applying Ampère's law along path C_1 lying inside the conductor, we have $\oint_{C_1} \mathbf{H} \cdot d\mathbf{l} = \int_{A_1} \mathbf{J} \cdot d\mathbf{s}$

$$\int_{\phi=0}^{2\pi} (H_\phi \mathbf{a}_\phi) \cdot (\mathbf{a}_\phi \rho_1 d\phi) = \int_0^{2\pi}\int_0^{\rho_1} K\sqrt{\rho}\mathbf{a}_z \cdot \mathbf{a}_z \rho d\rho d\phi = I_1$$

where I_1 is the portion of the total current I through the area $A_1 = \pi\rho_1^2$ enclosed by contour C_1. When $\rho_1 = a$, $I_1 = I$, this gives

$$\int_0^{2\pi}\int_0^a K\rho\sqrt{\rho}d\rho d\phi = I \quad \Rightarrow \quad K = \frac{5I}{4\pi a^{5/2}}$$

From the above we have

$$H_\phi \rho_1 \int_0^{2\pi} d\phi = 2\pi K \frac{2\rho_1^{5/2}}{5} \quad \Rightarrow \quad H_\phi = \frac{2K\rho_1^{3/2}}{5}, \quad \rho_1 \leq a$$

Similarly, by applying Ampère's law along contour C_2 lying outside the conductor (i.e., $\rho_2 > a$), we find

$$H_\phi\left(2\pi\rho_2\right) = I \quad \Rightarrow \quad H_\phi = \frac{I}{2\pi\rho_2}, \quad \rho_2 > a$$

In summary, the **H** field at any position ρ is

$$\mathbf{B} = B_\phi \mathbf{a}_\phi = \mu_0 \mathbf{H} = \begin{cases} \mu_0 \dfrac{I\rho^{3/2}}{2\pi a^{5/2}}\mathbf{a}_\phi, & \rho \leq a \\ \\ \mu_0 \dfrac{I}{2\pi\rho}\mathbf{a}_\phi, & \rho > a \end{cases}$$

Since

$$\mathbf{B} = B_\phi \mathbf{a}_\phi = \nabla \times \mathbf{A} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial\phi} - \frac{\partial A_\phi}{\partial z}\right]\mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right]\mathbf{a}_\phi + \frac{1}{\rho}\left[\frac{\partial\left(\rho A_\phi\right)}{\partial\rho} - \frac{\partial A_\rho}{\partial\phi}\right]\mathbf{a}_z$$

Comparing both sides, we get

$$B_\phi \mathbf{a}_\phi = \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right]\mathbf{a}_\phi$$

Keep the following relation in mind $\mathbf{A}(\mathbf{r}) = \int_V \frac{\mu_0 \mathbf{J} dv}{4\pi|\mathbf{r}-\mathbf{r}'|}$. So, $\mathbf{A}(\mathbf{r})$ has the same (or opposite) direction as that of $\mathbf{J}$ but its value depends on the field point $\mathbf{r}$. Since we consider an (almost) infinite straight conductor in the direction of z, it is obvious that the field $\mathbf{A}$ is in the direction of z, but does not change with z; rather like the current density $\mathbf{J}$, it is the function of ρ both inside and outside the conductor. So, we get

$$B_\phi = -\frac{\partial A_z}{\partial\rho} \quad \Rightarrow \quad \mathbf{A} = -\int B_\phi \, d\rho \, \mathbf{a}_z$$

giving

$$A_z = \begin{cases} -\mu_0 \dfrac{I\rho^{5/2}}{5\pi a^{5/2}} + c_1, & \rho \leq a \\ \\ -\mu_0 \dfrac{I\ln(\rho)}{2\pi} + c_2, & \rho > a \end{cases} = \begin{cases} -\mu_0 \dfrac{4K\rho^{5/2}}{25} + c_1, & \rho \leq a \\ \\ -\mu_0 \dfrac{I\ln(\rho)}{2\pi} + c_2, & \rho > a \end{cases} \tag{1}$$

where c_1 and c_2 are constants of integrations. An **alternate** way is to consider the fact that the magnetic vector potential satisfies Poisson's equation:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

In the cylindrical coordinate system, we have (this expression is not available in the textbook):

$$\nabla^2 \mathbf{A} = \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_\rho}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 A_\rho}{\partial \phi^2} + \frac{\partial^2 A_\rho}{\partial z^2} - \frac{2}{\rho^2}\frac{\partial A_\phi}{\partial \phi} \right) \mathbf{a}_\rho$$
$$+ \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_\phi}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 A_\phi}{\partial \phi^2} + \frac{\partial^2 A_\phi}{\partial z^2} + \frac{2}{\rho^2}\frac{\partial A_\rho}{\partial \phi} \right) \mathbf{a}_\phi$$
$$+ \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 A_z}{\partial \phi^2} + \frac{\partial^2 A_z}{\partial z^2} \right) \mathbf{a}_z = -\mu_0 \mathbf{J}.$$

Since $\mathbf{J} = K\sqrt{\rho}\,\mathbf{a}_z$ for $\rho \le a$, where K is a positive constant; comparing the coefficients of $\mathbf{a}_z$, we have

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 A_z}{\partial \phi^2} + \frac{\partial^2 A_z}{\partial z^2} = -\mu_0 K\sqrt{\rho}.$$

By appreciating the fact that A_z is not the function of z and ϕ, we obtain

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) = -\mu_0 K\sqrt{\rho}.$$

$$\rho \frac{\partial A_z}{\partial \rho} = -\mu_0 K \int \sqrt{\rho}\,\rho\, d\rho = -2\mu_0 K \frac{\rho^{5/2}}{5} + k_1$$

$$\frac{\partial A_z}{\partial \rho} = -2\mu_0 K \frac{\rho^{3/2}}{5} + \frac{k_1}{\rho}$$

$$A_z = -\mu_0 \frac{4K\rho^{5/2}}{25} + k_1 \ln(\rho) + k_2$$

. For $\rho > a$, $J_z = 0$, so the governing equation simplifies to:

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) = 0.$$

Multiply through by ρ:

$$\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) = 0.$$

Integrate with respect to ρ:

$$\rho \frac{\partial A_z}{\partial \rho} = k_3.$$

Divide through by ρ:

$$\frac{\partial A_z}{\partial \rho} = \frac{k_3}{\rho}$$

Integrate again with respect to ρ:

$$A_z = k_3 \ln\rho + k_4.$$

Boundary Conditions: To determine the constants, use the following boundary conditions:
1. Continuity of A_z at $\rho = a$, that is, $A_z(\rho \to a^-) = A_z(\rho \to a^+)$.
2. Continuity of B_ϕ, where $\mathbf{B} = \nabla \times \mathbf{A}$, that is, $B_\phi = -\frac{\partial A_z}{\partial \rho}$. The tangential component of $\mathbf{B}$ must also be continuous at $\rho = a$.

The value of $\mathbf{A}$ diverges when $\rho > a$ because $\ln(\rho)$ approaches infinity when ρ approaches infinity. Note that $\nabla \cdot \mathbf{A} = 0$. both inside and outside the conductor which is an essential condition. This is good to see that the two approaches yield identical results (when an appropriate choice of constants k_1, k_2, k_3, and k_4 is made; this is left for you to figure out).

Q3: [**15**] Consider a rectangular loop placed near a pair of infinitely long wires, cables 1 and 2, as depicted in Figure 2. The two wires carry currents I flowing in opposite directions, with the magnitude of the current increasing with time, $I(t) = Kt$, $K > 0$.

(a) [**10**] Determine the electromotive force (V_{ind}) induced in the loop (i.e., the potential difference measured across the small gap if the loop were broken) for the rectangular loop shown in Fig. 2.

(b) [**05**] Explain why the direction of the induced current is as shown in the figure and not in the opposite direction. Provide a detailed explanation, considering the roles of both cables 1 and 2.

Answer: The induced emf for the circuit is

$$V_{\text{ind}} = \frac{\mu_0 h K}{2\pi} \ln\left[\frac{a(b+a+w)}{(b+a)(a+w)}\right]$$

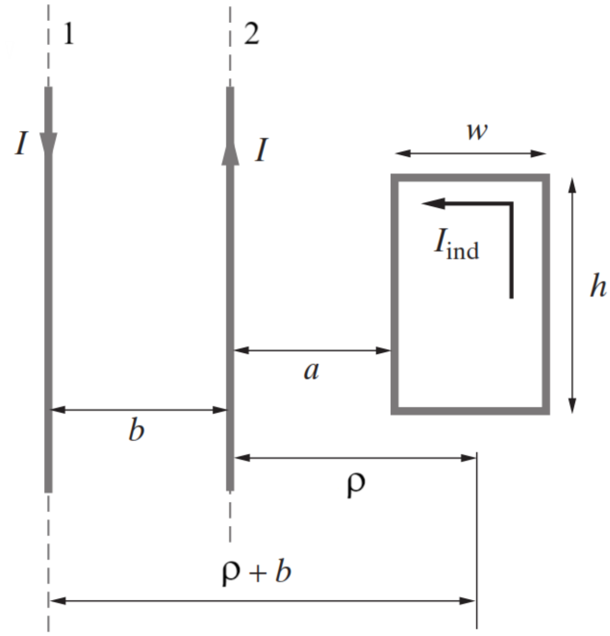

Figure 2: Pair of parallel wires and a loop.

Solution: We know that a current I in an infinitely long straight wire produces a field $\mathbf{B} = \hat{\phi} B_\phi$ at a distance ρ of $B_\phi = \frac{\mu_0 I}{2\pi\rho}$. Since the $\mathbf{B}$ field due to both infinitely long straight wires is given by the same expression, the total flux linking the loop can be written as

$$\Psi(t) = \frac{\mu_0 I}{2\pi} \int_{z=0}^{h} \left(\int_{\rho=a}^{a+w} \frac{d\rho}{\rho} - \int_{b+a}^{b+a+w} \frac{d\rho}{\rho} \right) dz = \frac{\mu_0 h I}{2\pi} \ln\left[\frac{(b+a)(a+w)}{a(b+a+w)}\right]$$

Q4: [**30**] Upon completing this course, you may consider applying your electromagnetics knowledge to real-world scenarios. Imagine a large power line passing near your locality, with your property 20 meters away from the line at ground level. You have 200 meters of copper wire available, which can be fashioned into a single-turn rectangular loop with dimensions a and b, as shown in Fig. 3. Assume that the power line carries a sinusoidal current with an amplitude of $I_0 = 4000\,\text{A}$ and a frequency of $f = 50\,\text{Hz}$. The instantaneous current is given by $I(t) = I_0 \cos(2\pi f t)$.

(a) [**10**] Considering the fixed perimeter of the loop: $l = 2a + 2b = 200$ m, show that the flux passing through the loop is:

$$\Psi(t) = \frac{\mu_0\, b\, I_0\, \cos(2\pi f t)}{2\pi} \ln\left(\frac{20+a}{20}\right)$$

(b) [**05**] Using Faraday's law, show that as

$$V_{\text{ind}}(t) = V_0 \sin(2\pi f t)$$

(c) [**05**] Find V_0, the amplitude of induced voltage, for b = 10, 20, 40, 60, 80 and 95 meters.

(d) [**10**] Plot V_0 versus b, and figure out the optimal value of b up to one decimal space that gives the largest value of V_0.

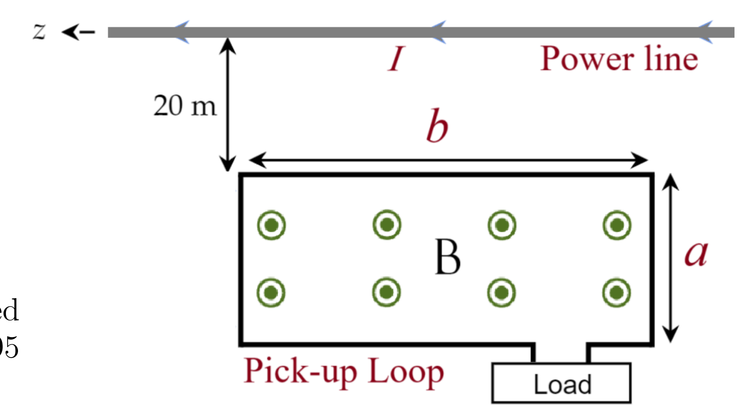

Figure 3: Extracting power from a power line.

(a) Recall the expression for the induced voltage, V_{ind} as defined by Faraday's Law $V_{ind} = -\frac{d\Psi}{dt}$ where we recall the expression for the magnetic flux linking the loop, $\Psi = \int_S \mathbf{B} \cdot d\mathbf{S}$, with the magnitude of Ψ which can be obtained for a differential surface/area dA where we note that the unit vectors for $\mathbf{B}$ and $d\mathbf{s}$ are parallel $\Psi = \int_S BdS$ which is the flux (Ψ) produced by the magnetic field $\mathbf{B}$ over the surface S. Moreover, for an infinitely-long conductor, its $\mathbf{B}$ field along $z = 0$ is defined by

$$\mathbf{B}_\phi \approx \frac{\mu_0 I}{2\pi\rho}\mathbf{a}_\phi$$

Solve for the expression of the flux Ψ

$$\begin{aligned}\Psi &= \int_S BdS = \int_{\rho=20\text{ m}}^{(20\text{ m})+a} \left(\frac{\mu_0 I}{2\pi r}\right) \cdot (bd\rho) \\ &= \frac{\mu_0 I(t) b}{2\pi} \int_{\rho=20}^{20+a} \frac{d\rho}{\rho} = \frac{\mu_0 I_0 \cos(2\pi f t)}{2\pi} b \ln\left(\frac{20+a}{20}\right)\end{aligned}$$

We note from the given, $2a + 2b = 200$ m $\rightarrow a = 100 - b$, and thus

$$\Psi = \frac{\mu_0 I_0 \cos(2\pi f t)}{2\pi} b \ln\left(\frac{120-b}{20}\right)$$

Solving for the induced voltage, we get

$$\begin{aligned}V_{\text{ind}} &= -\frac{d\Psi}{dt} = -\frac{d}{dt}\left[\frac{\mu_0 I_0 \cos(2\pi f t)}{2\pi} b \ln\left(\frac{120-b}{20}\right)\right] \\ &= -\frac{\mu_0 I_0}{2\pi} b \ln\left(\frac{120-b}{20}\right) \frac{d}{dt}[\cos(2\pi f t)] = -\frac{\mu_0 I_0}{2\pi} b \ln\left(\frac{120-b}{20}\right) \cdot 2\pi f[-\sin(2\pi f t)] \\ &= \mu_0 f I_0 \sin(2\pi f t) b \ln\left(\frac{120-b}{20}\right)\end{aligned}$$

We obtain an approximate value for the dimensions of our rectangular loop

$$\begin{aligned}b &\approx 61.919 \approx 62 \text{ m} \\ a &= 100 - b \approx 38 \text{ m}\end{aligned}$$